Unit ID: 6739e3b1a34577ff4ce151b8

unit 3 - Navigating Code Editor

PDF Documents

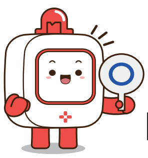
PDF Document 1: 1731847089176-Chapter 2 - unit 3 - Navigating Code Editor.pdf

This PDF document is included in the unit materials.

UNIT 3 NAVIGATING CODE EDITOR

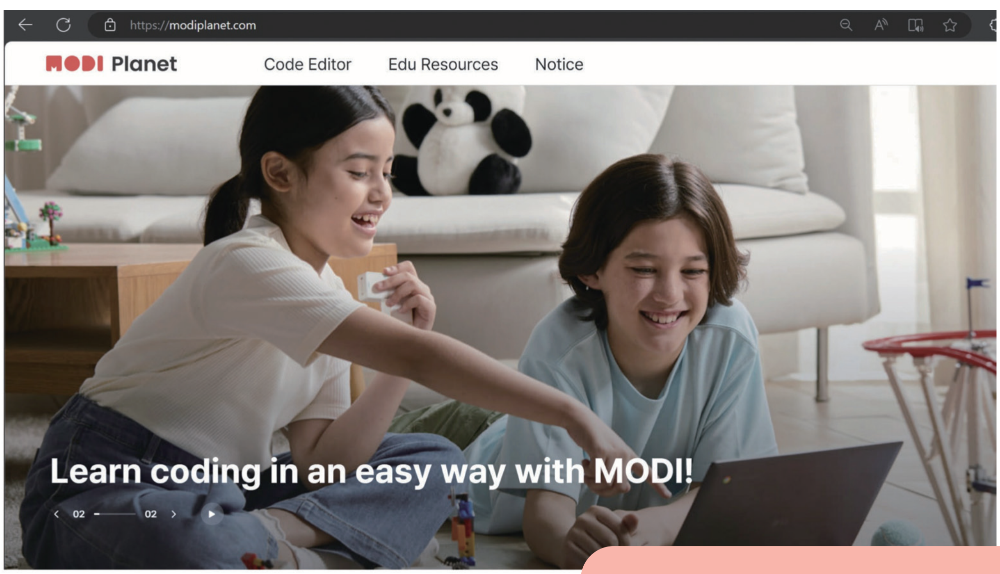
Introduction to Code Editor

What is Code Editor? Code Editor is where we can programm MODI modules with block-based programming.

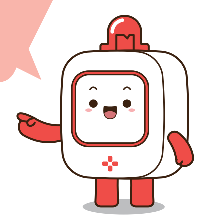


Let's get started by following the next steps:

Step 1 Go to 'MODI Planet' in your web browser. Here's the link: www.modiplanet.com

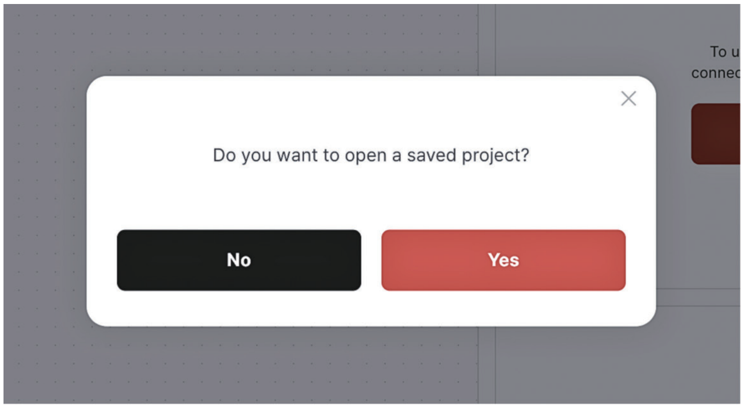


You can also search 'MODI planet' in a search engine!

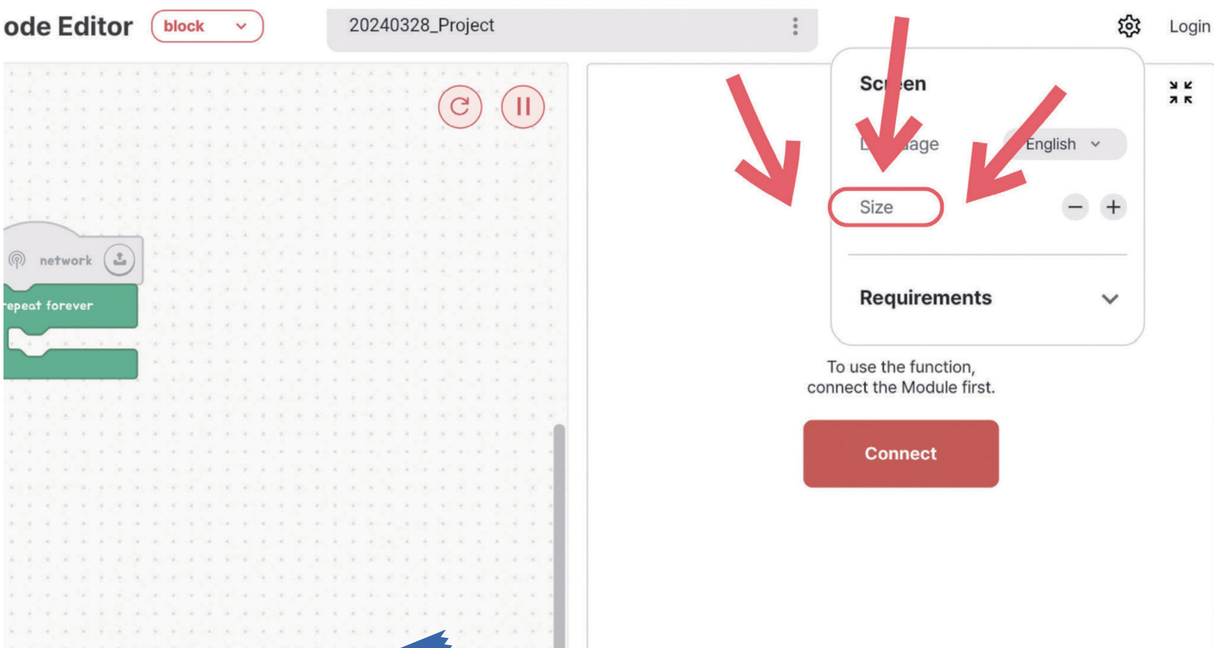


Step 2 Find 'Code Editor' in the top menu and click it.

Step 3 If it asks about opening a saved project, click 'No.'

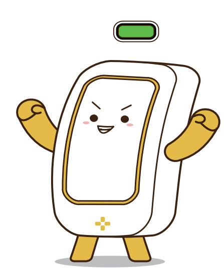


Step 4 If the screen layout doesn't suit your preference, adjust the ratio by clicking the "Set" button.

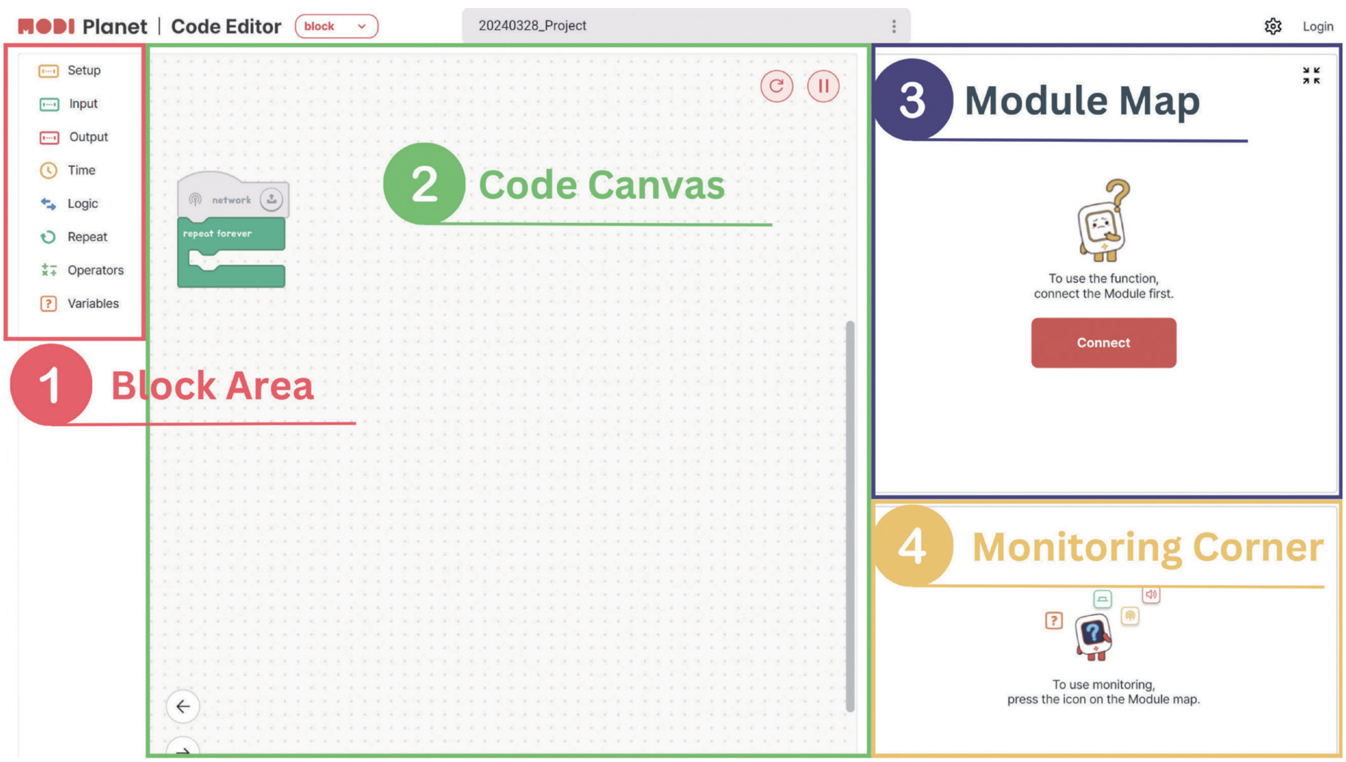


TIPS

Depending on your screen size, some options might look different. If you see a button in the top right corner, use it to see more options.



Exploring Code Editor



❶ Block Area :

This is where all the coding blocks live. They are sorted into eight groups based on what they do.

❷ Code Canvas :

This is your space to build your code by putting blocks together from the Block Area.

❸ Module Map :

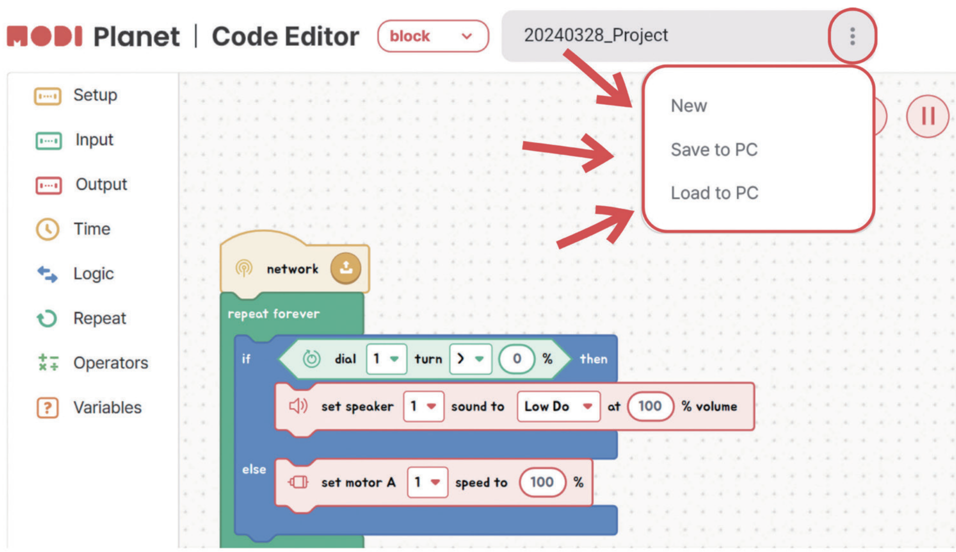
Here, you'll see icons for all the MODI modules connected to you PC. Clicking a module icon lets you check on that module.

❹ Monitoring Corner :

This is where the monitoring screen appears. Depending on the size of your screen, it may appear as a pop-up window.

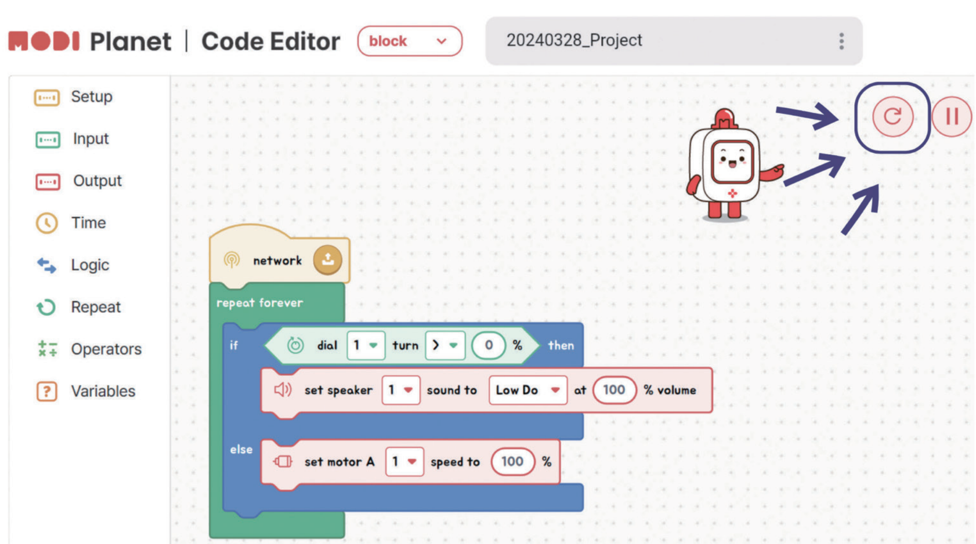
Save & Load Projects

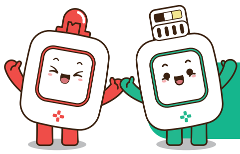
Click the ellipsis in the picture below to open a popup with options to save and load your projects.



Initializing modules

❶ Ensure the modules you wish to initialize are connected to your device.
❷ Click the 'Initialization' button to reset them.



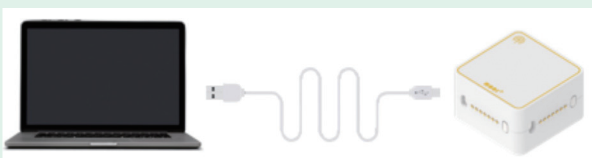


Connecting MODI to device

Once the MODI Code Editor is set up, let's connect your MODI modules to your device.

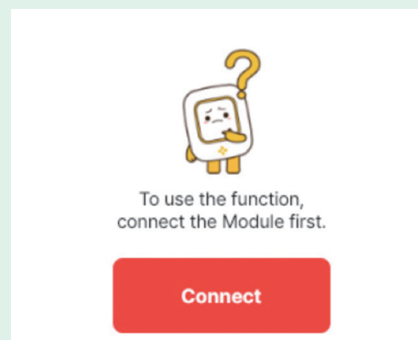
Step 1

Connect the Network Module to your device using a Type-C cable.



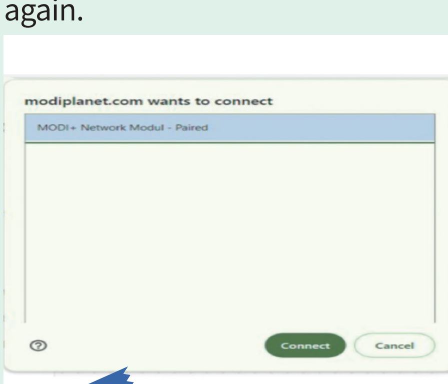
Step 2

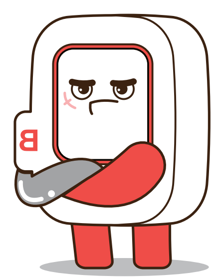
Click the connect button in the module map.



Step 3

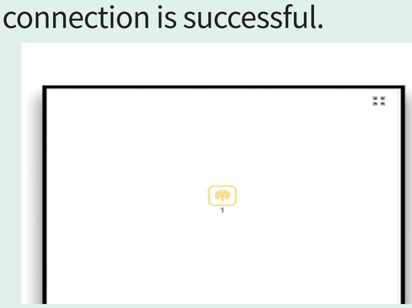
Select 'MODI+ Network Module-Paired' and then click 'Connect' again.

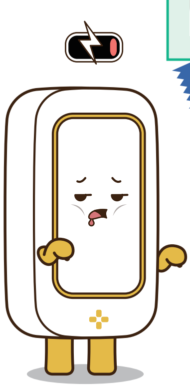




Step 4

A network icon will appear in the Module Map when the connection is successful.





TIPS

Any additional modules connected alongside the Network Module will be displayed in the Module Map.

Types of blocks in the block area

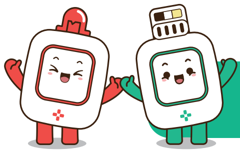
The Block Area is organized into seven tabs, each housing specific types of blocks:

Tab	No.	Description
Setup	1	**Network-Related Blocks:** These include the essential code start blocks that initiate your program.
Input	2	**Input Module Blocks:** Tailored for interacting with various input modules.
Output	3	**Output Module Blocks:** Designed to control and utilize output modules.
Time	4	**Time Block:** Used to pause the code execution for a set duration.
Logic	5	**Conditional Statements:** Includes 'if' and 'if-else' blocks for making decisions.
Repeat	6	**Loop Statements:** Contains blocks for repeating actions, like 'repeat'.
Operators	7	**Operators:** Focus on arithmetic calculations, comparisons, and generating random values.
Variables	8	**Variable Blocks:** Used for creating and modifying variables, crucial for storing and using data in your programs.

Blocks: Your First Step in Coding

Think of each block in the Code Editor as a piece of a puzzle that helps you build your program. Just like when you write instructions in a game or a recipe, blocks are the instructions you give to your computer.

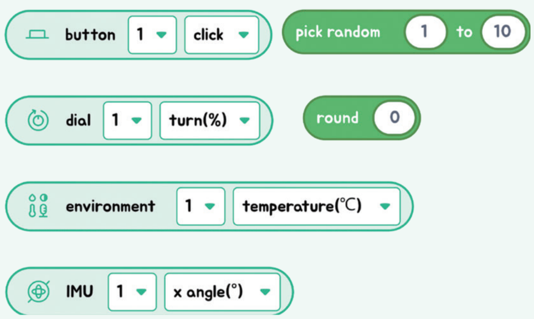
- **Network, Input, and Output Blocks:**
  - These blocks are like saying "go" in your program, telling it to start, take in information, or make something happen!
- **Time, Logic, and Loop Blocks:**
  - These blocks help decide what happens next, like choosing when to jump or keep running in a game.
- **Math and Variable Blocks:**
  - These blocks help you work with numbers and keep track of important data, like scores or clues.



Unplugged

In the Code Editor, there are three different shapes of blocks you'll use to create your programs. Each shape has a special job to do!

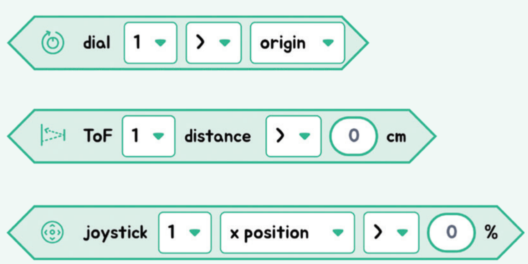
1



① Rounded Blocks:
These blocks hold values, like numbers. Think of them as small baskets carrying important information, usually numbers, for your code.

Examples:
- The button block tells when a button is clicked.
- The environment block can tell the temperature.

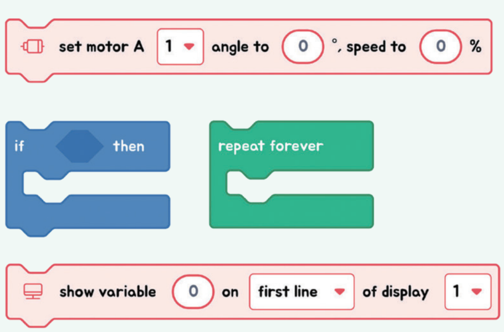
2



② Angled Blocks:
These blocks ask “yes” or “no” questions, dealing with conditions. If the answer is “yes” (true), one thing happens; if the answer is “no” (false), something else might happen.

Examples:
- The distance block asks, “Is this object close?”
- The joystick block asks, “Is the joystick pointing left?”

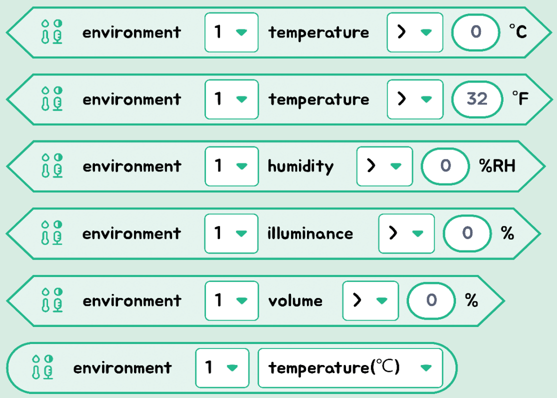
3

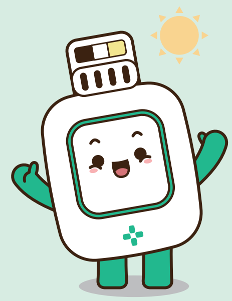


③ Rectangle-Shaped Blocks:
These blocks are all about action. They tell your MODI modules what to do, like moving, lighting up, or making sounds.

Examples:
- The motor block tells a motor to start turning.
- The show variable block displays information on a screen.

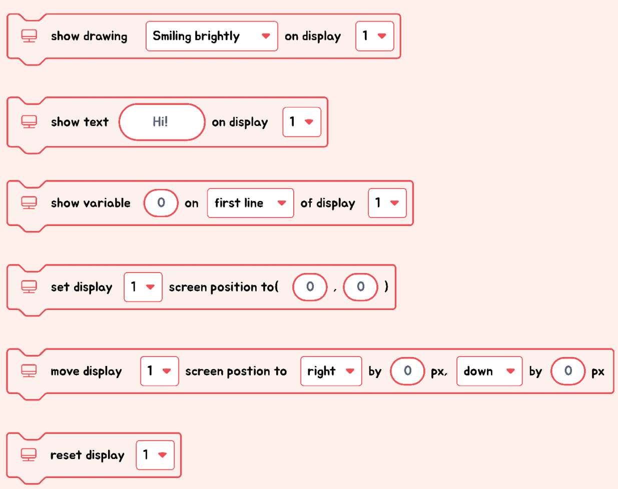
Blocks of Environment

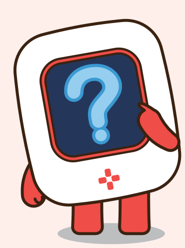




The input module called Environment doesn't perform actions. Instead, it provides 5 condition blocks and 1 value block, each related to different types of environmental data it can sense. The condition blocks are used to check specific environmental conditions, and the value block is used to retrieve the exact measurements of those conditions.

Blocks of Display

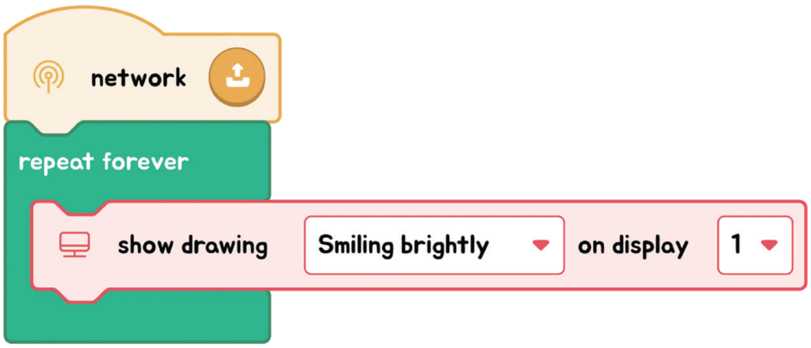




The output module, Display, is all about taking action. It has blocks specifically designed for showing pictures, text, or variable values on the screen. There are also blocks that allow you to move the displayed content around, and one important block that resets or initializes the Display for a fresh start.

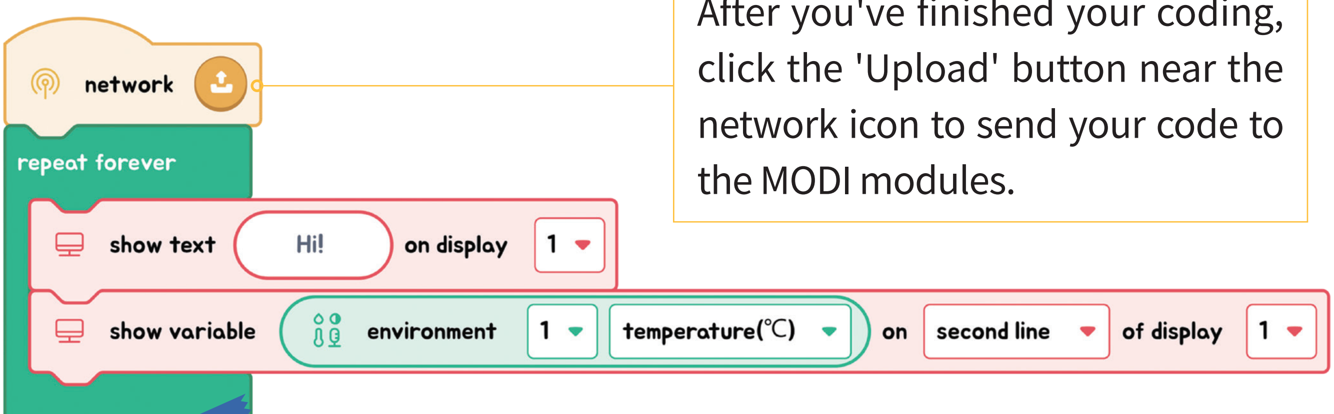
Showing a Picture on the Display

Let's get a picture to appear on the display screen. You can choose a different picture from the list and see how it looks!



Displaying Values on the Display

Now, let's have the display screen show the word 'temperature' and then, right below it, show the actual temperature that the Environment module detects.



After you've finished your coding, click the 'Upload' button near the network icon to send your code to the MODI modules.

TIPS

To prepare MODI for a new set of code, simply press the 'Initialize' button. You'll know MODI is ready when the status LED on the module turns blue.